

# JaeSeong Kim

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| <b>Research Interests</b>        | Computational imaging through joint optimization of optics, sensors, and algorithms. Differentiable ISP pipelines, polarization imaging, and end-to-end camera optimization for mobile, automotive, and microscopy applications.                                                                                             |                          |
| <b>Education</b>                 | <b>Tel Aviv University</b>                                                                                                                                                                                                                                                                                                   | Tel Aviv, Israel         |
|                                  | <i>M. Sc. in Electrical Engineering (94.19/100)</i><br>Advisor: Prof. David Mendlovic                                                                                                                                                                                                                                        | <i>April 2025</i>        |
| <b>Research Experience</b>       | <b>Tel Aviv University</b>                                                                                                                                                                                                                                                                                                   | Tel Aviv, Israel         |
|                                  | <i>B. Sc. in Electrical and Electronics Engineering (83.24/100)</i>                                                                                                                                                                                                                                                          | <i>July 2022</i>         |
| <b>Publication &amp; Patents</b> | <b>Tel Aviv University &amp; Samsung Corephotonics (Joint industrial Ms.c)</b>                                                                                                                                                                                                                                               |                          |
|                                  | <i>Research Assistant (with Prof. David Mendlovic)</i> <i>Oct 2022-Aug 2025</i>                                                                                                                                                                                                                                              |                          |
|                                  | <ul style="list-style-type: none"><li>• Polarization imaging for mobile face anti-spoofing. Developed method achieving 10x lower error rate than RGB solutions. Simulated hybrid sensor architectures and recruited 64 participants for novel polarization face dataset.</li></ul>                                           |                          |
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| <b>Employment</b>                | <b>Tel Aviv University</b>                                                                                                                                                                                                                                                                                                   |                          |
|                                  | <i>Research Assistant (with Prof. David Mendlovic)</i>                                                                                                                                                                                                                                                                       | <i>Jul 2021-Jul 2022</i> |
|                                  | Optics & Computer Vision Lab                                                                                                                                                                                                                                                                                                 |                          |
|                                  | <ul style="list-style-type: none"><li>• Developed CNN-based crime classification system using skeleton tracking to analyze human movements in surveillance videos, classifying actions into 10 crime categories.</li></ul>                                                                                                   |                          |
| <b>Publication &amp; Patents</b> |                                                                                                                                                                                                                                                                                                                              |                          |
|                                  | "On the Effectiveness of Sparse Linear Polarization Pixels for Face Anti-Spoofing", <b>J. Kim</b> , A. Pelz, M. Scherer and D. Mendlovic, <i>IEEE Sensors Journal</i> , 2025                                                                                                                                                 |                          |
|                                  |                                                                                                                                                                                                                                                                                                                              |                          |
|                                  | "Sensors, systems and methods for compact face identification polarization cameras", M. Scherer, <b>J. Kim</b> , Patent WO2024161329A1, 2024                                                                                                                                                                                 |                          |
| <b>Employment</b>                |                                                                                                                                                                                                                                                                                                                              |                          |
|                                  | "Enhanced Face Anti-Spoofing using Angle of Linear Polarization", <b>J. Kim</b> , M.Sc. Thesis, Tel Aviv University, 2025                                                                                                                                                                                                    |                          |
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| <b>Employment</b>                | <b>Samsung Corephotonics</b>                                                                                                                                                                                                                                                                                                 | Tel Aviv, Israel         |
|                                  | <i>Image Quality Engineer</i>                                                                                                                                                                                                                                                                                                | <i>Nov 2022-Apr 2025</i> |
|                                  | <ul style="list-style-type: none"><li>• Developed automated tool for quantifying EIS angular correction and motion blur, improving measurement accuracy by 25% and reducing analysis time from hours to 10 minutes</li><li>• Tuned ISP parameters using Qualcomm Chromatix and measured image quality with IMATEST</li></ul> |                          |
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|                               | <b>Pit-In Global</b> <span style="float: right;">Seoul, Korea</span><br><i>Software Developer (internship)</i> <span style="float: right;"><i>Jul 2017-Sep 2017</i></span><br><ul style="list-style-type: none"> <li>• Created SQL database for location tracking by developing software for Raspberry Pi that leverages Bluez to collect Bluetooth signal strength indicators (RSSI) from customers' mobile devices</li> </ul>            |
|                               | <b>CoderBunker</b> <span style="float: right;">Shanghai, China</span><br><i>Community Manager</i> <span style="float: right;"><i>Jan 2016-Mar 2016</i></span>                                                                                                                                                                                                                                                                              |
|                               | <b>Xinchejian(新车间)</b> <span style="float: right;">Shanghai, China</span><br><i>Community Manager</i> <span style="float: right;"><i>2015</i></span>                                                                                                                                                                                                                                                                                       |
| <b>Presentations</b>          | <b>Samsung Corephotonics Lecture Club</b><br>"Face Anti-spoofing using full polarization characteristics"<br>March 2025<br><br><b>Tel Aviv University Electrical Engineering Department Student Seminar</b><br>"Enhanced Face Anti-spoofing using Angle of Linear Polarization"<br>March 2025<br><br><b>Korean Embassy in Israel Research Seminar</b><br>"Camera Technologies - Multimodal cameras & Image Quality Assessment"<br>May 2024 |
| <b>Skills</b>                 | <ul style="list-style-type: none"> <li>• Technical Skills (Image Processing, Computer Vision, Deep Learning)</li> <li>• Programming (Python, MATLAB, C/C++)</li> <li>• Tools / Software (IMATEST, OpenCV, PyTorch, Qualcomm Chromatix)</li> </ul>                                                                                                                                                                                          |
| <b>Certifications</b>         | <ul style="list-style-type: none"> <li>• Oracle Certified Professional, JAVA SE 6 Programmer</li> <li>• Deep Learning Specialization (by Andrew Ng, deeplearning.ai)</li> </ul>                                                                                                                                                                                                                                                            |
| <b>Additional Information</b> | <ul style="list-style-type: none"> <li>• Military Service: KATUSA (Korean Augmentation to the United States Army), Republic of Korea Army (Oct 2017 – Jun 2019)</li> <li>• Korean (Native), English (Native), Chinese (Native), Hebrew (Beginner)</li> <li>• Multicultural background with experience living in Korea, China (11 years), and Israel (6.5 years)</li> </ul>                                                                 |